

Did COVID-19 restrictions in 2020 affect home field advantage in the NFL — controlling for week of play, team strength, and divisional matchups?

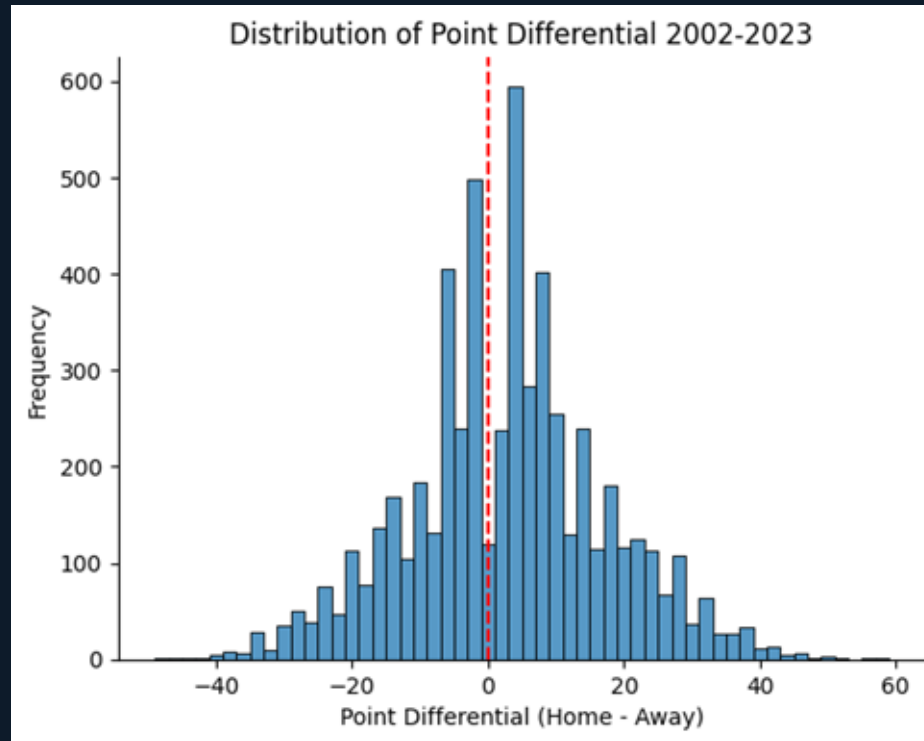
Aiden McCoy and Wyatt McMullen — ECON 150 Spring 2026

6,000+

REGULAR SEASON GAMES

2002–2023

SEASONS ANALYZED

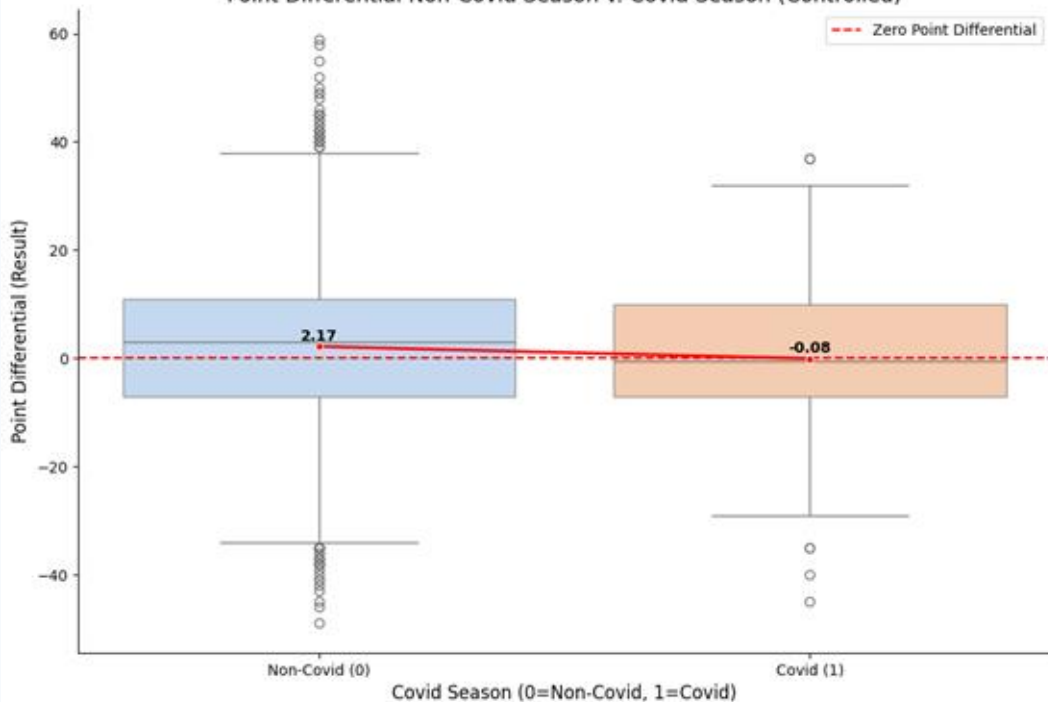


Point Diff > 0

HOME TEAM WINS!

MAIN RESULTS

Point Differential Non-Covid Season v. Covid Season (Controlled)



$$\text{Point Diff} = \beta_0 + \beta_1(\text{covid}) + \beta_2(\text{win_diff}) + \beta_3(\text{week}) + \beta_4(\text{div_matchup}) + \epsilon$$



Fans Matter

The results suggest the Covid-19 restrictions may have contributed to a near complete reduction in home field advantage, seen in the reduction of point differentials by 2.25 points on average ($p \approx 0.005$).

covid ≈ -2.25 pts | $p \approx 0.005$ | $R^2 \approx 0.2737$ (full model)